

Introduction to Data Science TASK 2

Muhammad Hassan Khalid

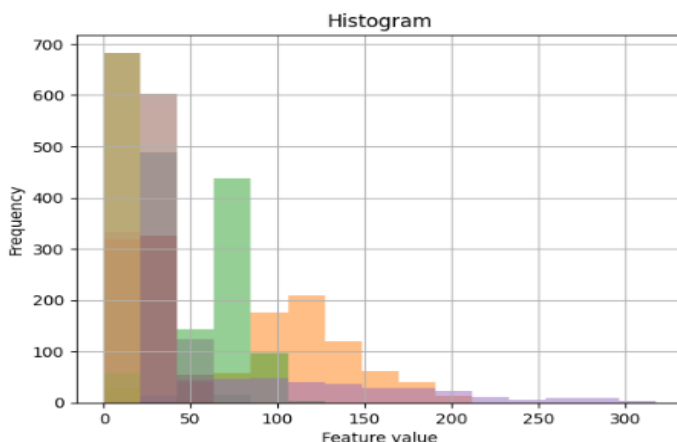
21L-5692

Dataset Using: - diabetes2.csv dataset

Dataset Source: - <https://www.kaggle.com/datasets/kandij/diabetes-dataset>

There are 9 dimensions of the dataset which include Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, Age and Outcome. I used the dimensions Glucose, BloodPressure, SkinThickness, Insulin for my data preprocessing and on the other dimensions including these applied logistic regression.

Firstly, I found all the values that had null values and replaced them by mean of the data given. Also, I detected the outliers in the dimensions Glucose, BloodPressure, SkinThickness, Insulin and omitted them from my data for a better understanding for my classifier. Then I plotted a bar graph for my data and found out that the data is right skewed also I calculated the variance for each dimension and then performed standard scaling on my data then I normalized the data to make my data cleaner and performed some further pre-processing steps for a cleaner data so that the classifier does a better job at learning the algorithm and making correct predictions.



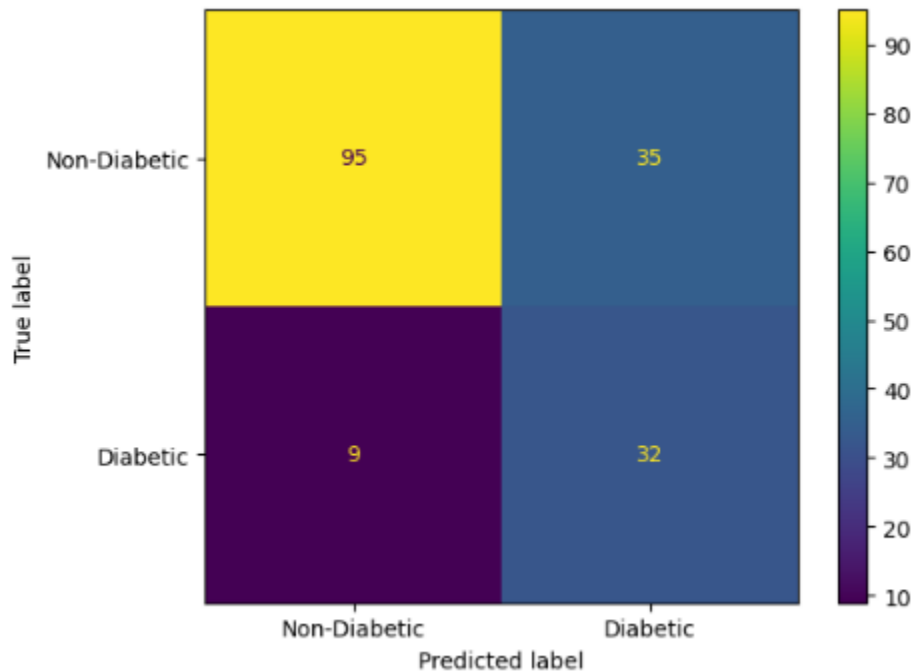
Pregnancies	0
Glucose	5
BloodPressure	45
SkinThickness	1
Insulin	34
BMI	0
DiabetesPedigreeFunction	0
Age	0
Outcome	0

dtype: int64

I have chosen Logistic Regression as my machine learning algorithm as it is binary form of outcome of data whether a person is diabetic or non-diabetic 0 or 1. Hence, logistic regression is better for my data.

My classifier did a good job in learning the data and out came a good classifier that was able to correctly classify the non-diabetic patients with an accuracy of 74%. It also did a quite good job in detecting the diabetics with a little false positives and negatives. Overall, the classifier was able to handle any value thrown at it and is at its final stages before being let out into the real world to make real decisions. Also I have attached below a confusion matrix below for your visual aid of the outcomes that came from which

it can be clearly seen the classifier was able to do a good job also the other things such as Recall, Precision , F1 Score and Accuracy are attached below for your reference and understanding.



```
In [351]: from sklearn.metrics import accuracy_score
print("Accuracy:" , accuracy_score(y_pred,y_test))
```

Accuracy: 0.7426900584795322

```
In [352]: from sklearn.metrics import precision_score, recall_score, f1_score
print("Recall:" , recall_score(y_test, y_pred,average='macro'))
```

Recall: 0.6955367393800229

```
In [353]: print("Precision: " , precision_score(y_test, y_pred,average='macro'))
```

Precision: 0.7556285178236397

```
In [354]: print("F1 Score: ", f1_score(y_test, y_pred,average='macro'))
```

F1 Score: 0.7022792022792022